Q15. Email header field investigation

Original question: Investigate a supplied email message using suitable Kali Linux tools. Extract and examine the sender, recipient, subject, date, message identifier, and return-path information. Record your observations.

Aim

Extract sender, recipient, subject, date, Message-ID and Return-Path from the supplied .eml and distinguish visible fields from transport evidence.

Requirements

Original exported .eml, Python 3 and terminal tools. Parsing does not render HTML or execute attachments. No message has been supplied here. Reference: [R15].

Procedure

Step 1 - Preserve and inspect raw headers

```
mkdir -p "$HOME/EHF_Exam/Q15"
cd "$HOME/EHF_Exam/Q15"
EMAIL='<EMAIL_FILE>'
sha256sum "$EMAIL" | tee email_before.sha256
file "$EMAIL"
sed -n '1,/^\r\{0,1\}$/p' "$EMAIL" | cat -v > raw_headers.txt
sed -n '1,100p' raw_headers.txt
```

A blank line separates the header section from the body; cat -v makes control characters visible. This is a quick raw view, not a complete MIME parser. Header folding means a field may continue on indented lines, so simple grep can omit important content. For reliable extraction use the complete script below.

Step 2 - Create and run a header parser

```
cat > inspect_headers.py <<'PY'
import sys
from email import policy
from email.parser import BytesParser
from email.utils import getaddresses

if len(sys.argv) != 2:
    raise SystemExit('Usage: python3 inspect_headers.py message.eml')
with open(sys.argv[1], 'rb') as stream:
    message = BytesParser(policy=policy.default).parse(stream)
fields = ('From', 'To', 'Cc', 'Reply-To', 'Subject', 'Date',
          'Message-ID', 'Return-Path')
for field in fields:
    values = message.get_all(field, [])
    print(f'{field}:')
    if not values:
        print('  [not present]')
    for value in values:
        print(' ', ascii(str(value)))
for field in ('From', 'To', 'Cc', 'Reply-To'):
    values = [str(v) for v in message.get_all(field, [])]
    print(field + ' parsed addresses:', ascii(getaddresses(values)))
print('Top-level parser defects:', ascii(message.defects))
PY
python3 inspect_headers.py "$EMAIL" | tee parsed_headers.txt
sha256sum -c email_before.sha256
```

BytesParser handles bytes, folded headers and standard encoded text. `get_all` retains duplicate fields for review. `ascii` escapes non-ASCII/control characters so header content is safer to inspect in a terminal; escaped text is deliberate, not corruption. The script extracts fields, not cryptographic authentication or an SMTP session log.

Expected output - illustrative

```
From: 'College Lab <lab@example.org>'
To: 'Student <student@example.net>'
Subject: 'EHF practical schedule'
Date: 'Sun, 18 Oct 2026 10:00:00 +0530'
Message-ID: '<lab-001@example.org>'
Return-Path: '<bounces@example.org>'
```

This is a compact synthetic summary, not the script's exact multiline formatting or a real email. Fields may be absent or duplicated. Record the literal results and timezone from your message.

Interpretation, observation and result

From is the claimed visible author; To/Cc are visible recipients, not a complete record of SMTP envelope recipients (Bcc often is absent). Subject and Date are author-supplied and can be misleading. Message-ID is intended to identify a message but can be forged, duplicated or missing. Return-Path normally records the envelope reverse-path inserted at final delivery; it may differ from From for legitimate bounce handling and can be empty for bounces. Its trust depends on provenance. Reply-To can direct responses elsewhere; a mismatch is a review point, not proof of phishing.

Record field/value/significance and any unusual duplicates, absent fields or address differences. Use Received and trusted authentication fields in Q17 to examine transport; do not infer the original sending IP from the From address. The result is an accurate field inventory with limitations, not sender attribution.

Common errors

- Input is Outlook .msg or a screenshot: obtain original .eml/full message export; renaming does not convert format.
- Grep misses a value: inspect folded lines or use the parser.
- Encoded/non-ASCII subject: the parser decodes standard encodings; retain raw headers alongside the normalized view.

Viva questions

- What separates headers from body? The first empty line.
- Is From the same as SMTP MAIL FROM? No; visible author and envelope return address serve different roles.
- Does Message-ID prove origin? No; it is not an authentication mechanism.
- Why preserve duplicate fields? They can indicate malformed or ambiguous message structure.